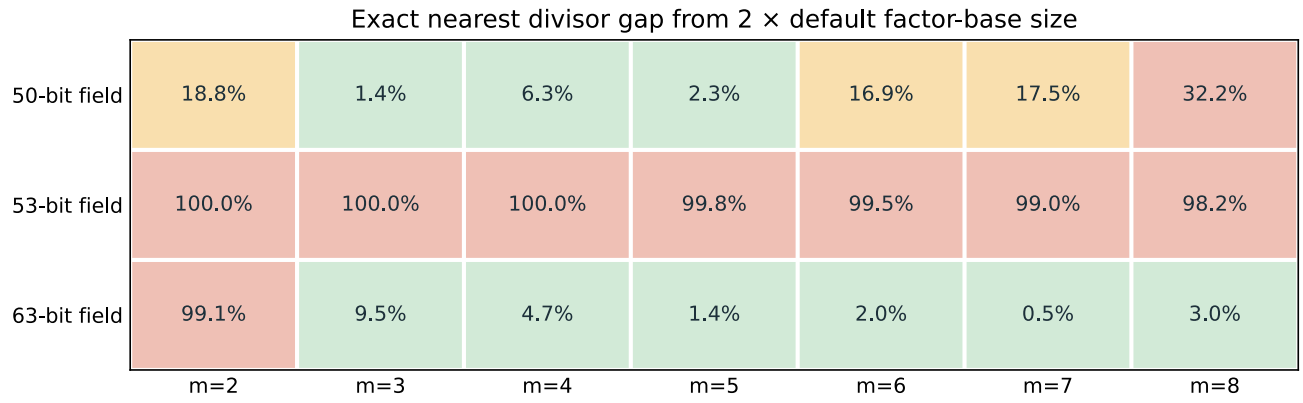
Prime-field divisor fit and a target-blind m=8 base

Input preparation and correctness observations, 2026-10-17 UTC. Exact factorizations of p−1 were checked for three certified public prime-order curves. The chart shows the closest divisor to twice the default factor-base size for m=2…8. Divisor fit is neither realized point count nor ECDLP speed.



Green: within 15%; amber: 15–25%; red: above 25%. Values rounded to 0.1%.
Source: research/icperf explicit divisor 20261017/divisor fit.json; field eligibility only.

The fixed 53-bit field has p−1 = 2 × 3 × 1501199800786417; its closest small divisor is 6 for every cell. This excludes a matched-size multiplicative coset base on that field, not on other 53-bit fields. The 63-bit field has a divisor within 15% of twice the default size for m=3…8.

Exact m=8 preparation on the 63-bit field

Quantity	Recorded value
Default size B	406
Chosen divisor d	788; relative divisor gap 24/812 = 2.96%
Realized base	380 distinct on-curve x-coordinates
Membership equation	x^788 − g^788; two nonzero terms
Root proof	ζ has exact order 788; one root draw

The base uses fixed seed 20261017 and was created before any target was chosen. Its complete point list and digest are in the machine-readable record. The checker independently enumerated all 788 coset x-values, lifted each one, compared the ordered 380-point list, and rejected a changed-point control. The explicit-divisor code avoids factoring all of p−1 during base construction; the field-divisor selection itself used a Sage factorization and remains setup work to charge in a future trial.

Remaining scientific gate

An equal-realized-size 380-point random and small-x control, planted and unplanted decomposition, direct S8 capacity, eight-summand S9 or chained S3, observed solver degree, useful relation rank, complete target work and matched rho are unmeasured. A sparse input equation does not establish low degree of regularity. The archived 0.053-second constructor timing excludes interpreter startup, field search and factorization and is not a complete cost.

The canonical ECDLP frontier graph remains unchanged; this search adds no reviewed attack-cost result. Source paths and exact digests are in the companion Markdown report.